Lecture 10. Family-wise Error Rate Control

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1. Mutiple Testing / Comparison Problems

Until now, we have been considering tests of the global null $H_0 = \bigcap_i H_{0,i}$. For some testing problems, however, our goal is to accept or reject each individual $H_{0,i}$ instead of the global null while control the overall type I and type II error.

Motivative Example: DNA microarrays measure expression levels of tens of thousands of genes. The data consist of levels of mRNA, which are thought to measure how much of a protein the gene produces. A larger number implies a more active gene.

$$X_{ij} = \text{expression level of gene } i \text{ in patient } j.$$

Here we have n tests of hypotheses:

$$\begin{aligned} &H_{0,i} : \text{gene } i \text{ is ``null''} \\ &H_{1,i} : \text{gene } i \text{ is ``non-null''} \end{aligned}$$

and we are interested in calling each hypothesis $1 \le i \le n$. In this case the global null could still be interesting if we were trying to discover whether prostate cancer has a genetic component.

We collect p-values $p_1, \cdots, p_n$ for each test. In our example,

$$T_i = \frac{\text{Ave[Cancer] -Ave[Control]}}{\text{Est. Std. Error}}$$

Under $H_{0,i}, T_i \sim t_{m-2}$ with m being the number of patients, and the corresponding p-values are $p_i = \mathbb{P}(|t_{m-2}| > |T_i|)$.

Generally, we have four types of outcomes in multiple testing:

	accept	rejected	total
True	U	V	n_0
False	T	S	$n - n_0$
total	$n - R$	R	n

where U, V, S, T are unobserved random variables but R is observed.

The two quantities of primary interest to us are the size of V (the number of false discoveries) andthe size of R (the total number of rejected hypotheses).

Definition 1.1 (♣ **Family-wise Error Rate (FWER)**). Family-wise error rate is the criterion used commonly in classical multiple comparison procedures aim to control

$$\text{FWER} = \mathbb{P}(V \geq 1)$$

in a strong sense, that is, under all configurations of true and false hypotheses.

2. Procedures for Controlling FWER

Methods for controling the global null usually works for controling the FWER.

2.1. Bonferroni's Approach

Recall that Bonferroni's approach reject $H_{0,i}$ for which $p_i \leq \alpha/n$. Therefore

Theorem 2.1 (♣ **Controling FWER According to Bonferroni's Approach**). *Bonferroni's approach controls FWER at level α in the strong sense, in fact,*

$$\text{FWER} \leq \mathbb{E}V = \frac{n_0 \alpha}{n}.$$

Proof. We introduce the random variables U_i, V_i, S_i, T_i, $1 \leq i \leq n$, where U_i is the indicator of we accpet $H_{0,i}$ when $H_{0,i}$ is indeed the truth, and V_i, S_i, T_i are similarly defined. Apparently, for each $1 \leq i \leq n$, there is one and only one variable of U_i, V_i, S_i, T_i takes value 1 and rest of them taks value 0. Denote the set of nulls by $\mathcal{H}_0$. Then $V = \sum_{i \in \mathcal{H}_0}(V_i)$ and

$$\mathbb{E}V = \sum_{i \in \mathcal{H}_0} \mathbb{E}(V_i) = \sum_{i \in \mathcal{H}_0} \mathbb{P}(V_i = 1) = \sum_{i \in \mathcal{H}_0} \frac{\alpha}{n} = \frac{n_0}{n}\alpha.$$

This concludes the proof since

$$\mathbb{P}(V \geq 1) \leq \mathbb{P}(V \geq 1) + \mathbb{P}(V \geq 2) + \cdots + \mathbb{P}(V \geq n) = \mathbb{E}V.$$

□

2.2. Sidak's Approach

Recall that Sidak's procedure requires independence between different hypothesis. Under independence, Sidak's procedure reject $H_{0,i}$ for which $p_i \leq \alpha_n$, and

$$\alpha_n = 1 - (1 - \alpha)^{1/n}.$$

Theorem 2.2 (♣ **Controling FWER According to Sidak's Approach**). *Sidak's approach controls FWER at level α in the strong sense, in fact,*

$$\text{FWER} = \mathbb{P}(V \geq 1) = 1 - \mathbb{P}(V = 0) = 1 - (1-\alpha)^{n_0/n}.$$

Proof. We adopt the notation used above, U_i, V_i, S_i, T_i, $1 \leq i \leq n$. Then $V = \sum_{i \in \mathcal{H}_0}(V_i)$ and

$$\mathbb{P}(V \geq 1) = \sum_{i \in \mathcal{H}_0} \mathbb{P}(V_i \geq 1) = 1 - \mathbb{P}(V_i = 0, i \in \mathcal{H}_0) = 1 - \mathbb{P}(V = 0) = 1 - (1-\alpha)^{n_0/n}.$$

Since $n_0/n \leq 1$, so

$$\mathbb{P}(V \geq 1) = 1 - (1-\alpha)^{n_0/n} \leq 1 - (1-\alpha) \leq \alpha.$$

Therefore, just for the purpose of control FWER, we may even set the critical value at

$$\alpha_n = 1 - (1-\alpha)^{1/n_0}$$

or a simpler version, by asking

$$\mathbb{P}(V \geq 1) = 1 - (1-\alpha_n)^n = 1 - 1 + n\alpha_n - \binom{n}{2}\alpha_n^2 + \cdots \leq \alpha.$$

In other words, we may just set

$$n\alpha_n\left(1 - \frac{n-1}{2}\alpha_n\right) \leq \alpha = 0.05$$

So for n large ($n \geq 10$, say), this is about the same as

$$n\alpha_n\left(1 - \frac{n\alpha_n}{2}\right) \leq 0.05$$

Rearranging, we obtain

$$\alpha_n \leq \frac{0.05}{n}\left(\frac{1}{1-0.025}\right) \approx \frac{0.05}{n}$$

Thus for even moderate values of n, using Sidak's procedure is effectively the same as using Bonferroni. It is not surprising that we cannot do better using Sidak's procedure for large n, since Bonferroni is optimal in a minimax sense for large samples. □

2.3. Holm's procedure

Holm's procedure is a step-down procedure operating as follows:

(i) first, we order the p-values as

$$p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(n)}$$

and let $H_{(1)}, H_{(2)}, \cdots, H_{(n)}$ be the corresponding hypotheses.

(ii) Examine the p-values in the following order.

(a) If $p_{(1)} \leq \alpha/n$ reject $H_{(1)}$ and go to Step 2. Otherwise, accept $H_{(1)}$, $H_{(2)}$, $\cdots$, $H_{(n)}$ and stop.

(b) If $p_{(i)} \leq \alpha/(n-i+1)$ reject $H_{(i)}$ and go to Step $i+1$. Otherwise, accept $H_{(i)}, H_{(i+1)}, \cdots, H_{(n)}$ and stop.

(c) If $p_{(n)} \leq \alpha$ reject $H_{(n)}$. Otherwise, accept $H_{(n)}$.

Hence the procedure stops the first time $p_{(i)}$ exceeds the critical value $\alpha_i = \alpha/(n-i+1)$.

Holm's procedure is not as conservative as Bonferroni; we typically make more rejections (have more power). Also Holm's procedure can always be used instead of Bonferroni, due to the following result.

Theorem 2.3 (♣ **Controling FWER According to Holm's Approach**). *Holm's procedure controls the FWER strongly.*

Proof. Let i_0 be the rank of the smallest null p-value so that in Holm's procedure, the first true null is encountered at step i_0. Obviously,

$$i_0 \leq n - n_0 + 1$$

i.e., the first true null can be preceded by at most $n - n_0$ other hypotheses (all the false ones). Holm's commits a false rejection if and only if

$$p_{(1)} \leq \frac{\alpha}{n}, \ \ p_{(2)} \leq \frac{\alpha}{n-1}, \ \ \cdots, \ \ p_{(i_0)} \leq \frac{\alpha}{n-i_0+1},$$

which implies

$$p_{(i_0)} \leq \frac{\alpha}{n-i_0+1} \leq \frac{\alpha}{n_0}.$$

Therefore the probability of false rejection is bounded above by

$$\mathbb{P}(\min_{i \in \mathcal{H}_0} p_i \leq \frac{\alpha}{n_0}) \leq \sum_{i \in \mathcal{H}_0} \mathbb{P}(p_i \leq \alpha/n_0) = \alpha$$

□

3. Global Testing vs. Multiple Testing

For global testing, we have p-values $p_1, \cdots, p_n$ and a global null $H : p_1, \cdots, p_n \sim$Uniform$[0,1]$. Our goal is to get α-level control under this global null: $\mathbb{P}_H[\text{reject } H] \leq \alpha$. We saw several procedures that can be used to test H; as usual, we denote the ordered p-values as $0 \leq p_{(1)} \leq \cdots \leq p_{(n)} \leq 1$.

- Bonferroni: Reject H if $p_{(1)} \leq \alpha/n$.
- Simes: Reject H if $p_{(1)} \leq \alpha/n$ or $p_{(2)} \leq 2\alpha/n$ or $\cdots$ or $p_{(n)} \leq \alpha$.
- Fisher: Reject H if $\sum_{i=1}^{n} 2logp_i^{-1} \geq X_{2n}^2(1-\alpha)$.

Now, for multiple testing, we have the same p-values as before, except we now care about the individual hypotheses $H_i : p_i \sim$Uniform$[0,1]$, for $i = 1, 2, \cdots, n$ separately. If each p-value tests the association between a gene expression level and a disease, then global testing asks whether any gene is associated with the disease, whereas multiple testing asks whether this gene is associated with the disease, for each gene separately.

Our goal is to control the family-wise error rate (FWER). To do so, define $\mathcal{M}^0 = \{i : H_i \text{ is true}\}$, and $\mathcal{R} = \{i : H_i \text{ is rejected}\}$. The FWER is defined as

$$\text{FWER} = \mathbb{P}[\mathcal{M}^0 \cap \mathcal{R} \neq \emptyset].$$

We want procedures for which FWER $\leq \alpha$. Some first observations:

- Bonferroni, i.e., reject H_i if $p_{(i)} \leq \alpha/n$, controls FWER "out of the box".
- Simes, i.e., reject $H_{(i)}$ if $p_{(j)} \leq j\alpha/n$ for some $j \geq i$, does not control FWER.
- Fisher's two-step procedure, i.e., reject $H_{(i)}$ if Fisher's test rejects the global null and $p_i \leq \alpha$, does not control FWER.

Thus, FWER control does not follow trivially from global control.

4. Closure Principle

Consider a family of hypotheses $\{H_i\}_{i=1}^n$. For indices i and j, we can define the intersection null

$$H_{ij} = H_i \cap H_j = \{p_i, p_j \sim U[0,1]\}.$$

More generally, define the closure of this family as

$$H_I = \bigcap_{i \in I} H_i \quad \text{for all } \ I \subset \{1, 2, \cdots, n\}.$$

For example, consider the case $n = 3$. The closure in this case is just

$$H_{123}, \quad H_{12} \ H_{13} \ H_{23}, \quad H_1 \ H_2 \ H_3.$$

For each I, take a α-level test φ_I for testing H_I (rejects if $\varphi_I = 1$)

$$\mathbb{P}(\varphi_I = 1 | H_I) \leq \alpha.$$

Notice that H_I is the "global null' for the index-set I. Thus, the tests φ_I may be generated by any global testing procedure (e.g., Bonferroni, Simes).

The Closure procedure: Reject H_I iff for all $J \supseteq I$, H_J is rejected at level α; that is, $T_I = min_{J \subseteq I} \Phi_J$. For instance, consider the case of 4 hypotheses. Suppose the underlined hypotheses are rejected at level α. Then the closure is

$$\underline{H_{1234}}, \quad \underline{H_{123}} \ \underline{H_{123}} \ \underline{H_{134}} \ H_{234},$$
$$\underline{H_{12}} \ \underline{H_{13}} \ \underline{H_{14}} \ H_{23} \ H_{24} \ H_{34}, \quad \underline{H_1} \ \underline{H_2} \ H_3 \ H_4.$$

In this example, only H_1 is rejected by the closed test procedure. H_2 is not rejected, although it is marginally significant (because not all intersections are found to be significant).

Theorem 4.1. *The closure principle controls the FWER strongly.*

Proof. Let $\mathcal{M}^0$ be the set of true nulls, and assume it's non-empty (otherwise no rejection can be false).

$$A : \{\text{at least one false rejection}\}$$
$$B : \{\text{reject } H_{\mathcal{M}^0}\}$$

For the closure procedure, $A \bigcap B = A$. So

$$\mathbb{P}(A) = \mathbb{P}(A \bigcap B) = \mathbb{P}(B)\mathbb{P}(A|B) \le \alpha.$$

□

Thus, the closure procedure provides us with a generic recipe for translating global testing procedures into valid FWER controlling procedures. Thus raise the question of whether the closure principle yields powerful multiple testing procedures. As it turns out, it often does (but not always).

4.1. Closing Bonferroni

$H_{0I}: \bigcap_{i\in I} H_{0i}$ v.s. $H_{1I}: \bigcup_{i \in I} H_{1i}$

We use Bonferroni's procedure to generate the φ_I:

$$\varphi_I = 1 \ \text{ iff } \inf\{p_i : i \in I\} \le \frac{\alpha}{|I|}.$$

one of the $p_i \le \frac{\alpha}{|I|}, i \in I$ $\Leftrightarrow$

We know that applying the closure principle to these test statistics will yield a valid FWER controlling procedure. However, evaluating the closure tests explicitly requires computing 2^n test statistics; ideally we would like to have an analytic short-cut for evaluating the closure. It turns out that the closure of the Bonferroni procedure has a simple closed-form solution.

To derive a concise description of the procedure, note the following:

$I \subseteq I_{(j)+}$

(i) Consider an index set I for which $p_{(j)} = \min\{p_i : i \in I\}$ where the subscript (j) denote this is the j-th smallest p-value. Let $I_{(j)+}$ denote the index set corresponding to the $n-j+1$ largest p-values $\{(j),(j+1),\cdots,(n)\}$. Then

$$\Phi_{I^+_{(j)}} = 1 \ \text{ implies that } \ \Phi_I = 1$$

Proof. Because $\Phi_{I^+_{(j)}} = 1$, we know that $p_{(j)} \le \alpha/(n-j+1)$. Now, because $p_{(j)}$ is the smallest p-value with index in I, we know that $|I| \le n-j+1$. Thus $p_{(j)} = \min\{p_i : i \in I\} \le \alpha/|I|$. □

(ii) Given the first observation, we notice that

$$\begin{aligned} H_{(j)} \text{ is rejected } &\Longleftrightarrow \phi_{I^+_{(1)}} = 1 \quad \text{and} \quad \cdots \quad \text{and} \quad \phi_{I^+_{(j)}} = 1 \\ &\Longleftrightarrow p_{(1)} \le \frac{\alpha}{n} \quad \text{and} \quad \cdots \quad \text{and} \quad p_{(j)} \le \frac{\alpha}{n-j+1}. \end{aligned}$$

$\Rightarrow$ Holms procedure

$I^+_{(j)} = \{(j), (j+1), \cdots (n)\}$

$|I^+_{(j)}| = n-j+1$

$|I| \le |I^+_{(j)}| = n-j+1$

$p_{(j)} = \min\{p_i, i \in I\} \le \frac{\alpha}{|I|}$

Initialize $j = 0$;
while $p(j+1) \leq \frac{\alpha}{n-j}$ **do**
  Increment j by 1;
end
Result: Reject hypotheses $H(1), \ldots, H(j)$

Algorithm 1: Procedure Closed Bonferroni

We can also describe the closed Bonferroni procedure procedurally, as above. Closing Bonferroni has thus led us back to a procedure we already knew: Holm's procedure. This is encouraging, in that the closure principle appears to yield reasonable procedures.

The closure principle does not always yield such nice procedures, e.g., closing Fisher's procedure.

4.2. Closing Simes

Suppose we now apply the closure principle with Simes test statistics

$$\varphi_I = 1(\{p_{(1,I)} \leq \alpha/|I|\}) \quad \textbf{or} \quad 1(\{p_{(2,I)} \leq 2\alpha/|I|\}) \quad \textbf{or} \quad ... \quad \textbf{or} \quad 1(\{p_{(|I|,I)} \leq \alpha\}),$$

where $p(j, I)$ denotes the j-th p-value with index in I. Because Simes is strictly more powerful thanBonferroni, the closure of Simes' procedure will be strictly more powerful than Holm's procedure.However, just like Simes, the closure of Simes requires the p-values to be independent (or for a similar weaker condition to hold).

This closure can be evaluated, although it is a little complicated. Here, we will derive a simple procedure that is strictly more conservative than the closure of Simes. Thus, because the closure of Simes controls FWER under independence, our procedure will also control FWER under independence.

Lemma 4.2. *For a set I, suppose that (a) the index (j) corresponding to the j-th smallest p-value is in I, and (b) that there exists a $j' \geq j$ such that $p_{(j')} \leq \frac{\alpha}{n-j'+1}$. Then $\varphi_I = 1$ for the test statistic φ_I given by Simes' procedure.*

Proof. Let k be the index given by

$$k = \sup\{i : (i) \in I \text{ and } i \leq j'\};$$

by hypothesis, we know that the set over which we are maximizing is non-empty. Then

$$\begin{aligned} p_{(k)} &\leq p_{(j')} \\ &\leq \frac{\alpha}{n-j'+1} \quad \text{(critical value in Simes)} \\ &\leq \frac{\alpha}{1+|\{(j'+1), ..., (n)\} \cap I|} \end{aligned}$$

$n-j'+1 = 1+|\{(j'+1), \ldots, (n)\}|$
$\geq 1+|\ldots \cap I|$
$= |\{(k), \ldots, (n)\} \cap I|$

$$= \frac{\alpha}{|\{(k), ..., (n)\} \cap I|}$$

$$\leq \frac{|\{(1), ..., (k)\} \cap I|}{|I|}\alpha$$

$\leq \frac{1+|\{(1),--,(k-1)\}|}{1 \quad 1+|\{(1),--,(k-1)\}|} \alpha$

$\frac{a}{b} < 1$, $\frac{a}{b} < \frac{a+c}{b+c}$

Thus, by definition of Simes' procedure, $\varphi_I = 1$. □

By closing Simes, we have verified that the following procedure controls FWER:

$$\text{Reject } H_{(j)} \text{ if there is an index } j' \geq j \text{ such that } p_{(j')} \leq \frac{\alpha}{n - j' + 1}.$$

This procedure is known as Hochberg's procedure.

5. Step-Down vs. Step-Up Procedures

We can write Holm's and Hochberg's procedures side-by-side:

Initialize $j = 0$;
while $p(j+1) \leq \frac{\alpha}{n-j}$ **do**
| Increment j by 1;
end
Result: Reject hypotheses $H(1), \ldots, H(j)$

Algorithm 2: Holm

Initialize $j = n$;
while $p(j) \geq \frac{\alpha}{n-j+1}$ **do**
| Decrement j by 1;
end
Result: Reject hypotheses $H(1), \ldots, H(j)$

Algorithm 3: Hochberg

The two procedures have the same thresholds, i.e., $p_{(j)}$ is compared to $\frac{\alpha}{n-j'+1}$. However,

1. Holm's scans forward, and stops as soon as a p-value fails to clear its threshold. This pessimistic approach is called a step-down procedure (think stepping downwards on the X^2-statistics).
2. Hochberg's scans backwards, and stops as soon as a p-value succeeds in passing its threshold.This optimistic approach is called a step-up procedure (think stepping upwards on the X^2-statistics).

In general, step-up procedures can be substantially more powerful than step-down procedures. In the extreme case where $p_1 = \cdots = p_n = \alpha$, Holm's rejects nothing whereas Hochberg's rejects everything. A perhaps less extreme example is shown in the following gure; here, by scanningforward, Holm's can only reject 3 hypotheses whereas Hochberg's can reject 8 hypotheses because $p_{(8)} \leq \alpha/(n - 8 + 1)$.

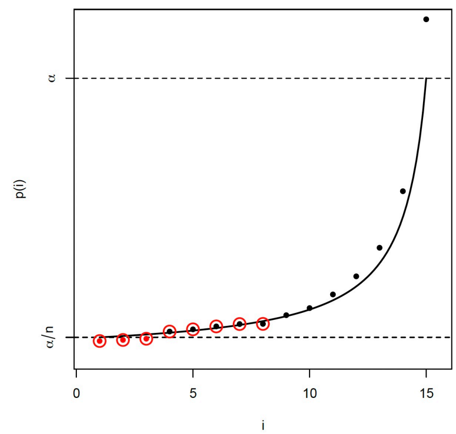

Figure 1: Comparison of Holm's Step-Down procedure and Hochberg's Step-Up procedure. Bothcontrol the FWER, but Hochberg's procedure is more powerful (makes more rejections). Solid redpoints are hypotheses rejected by Holm's procedure (3 rejections). Points circled in red are rejectedby Hochberg's procedure (8 rejections)

References

Williams, D. (1991). Probability with martingales. Cambridge university press.